

Matthias Druhl

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Education

Georgia Institute of Technology

B.S. Computer Science | Threads in Theory & Intelligence

Atlanta, GA

Graduation: May 2027

Experience

Georgia Institute of Technology | Academic and Research Computing Services

Atlanta, GA

Web Developer

November 2025 - Present

- Managed and optimized a fleet of 300+ institutional web applications using Drupal and Plesk, implementing performance tuning and load balancing to maintain a 99.997% service availability across the College of Sciences network
- Directed the successful redesign and migration of 40+ high-traffic web assets to modern CMS environments, automating data transfer protocols and ensuring cross-platform compatibility
- Resolved technical issues within a large-scale enterprise environment, leveraging root cause analysis to decrease recurring software bugs and improving operational workflows for 25k+ campus users

United States Air Force | Extreme Digital Development Group Enterprise

Macon, GA

Software Engineering Intern - Team Lead

May 2025 - July 2025

- Administered development of an end-to-end web application (Docker, React, FastAPI), automating the generation of virtual microcontroller emulators from datasheet PDFs, compressing a year-long manual development cycle into three months
- Designed and implemented a RAG-based LLM pipeline (LangChain/Ollama) that leveraged advanced PDF parsing and vector retrieval to transform unstructured datasheet data into virtual emulator code, improving code functionality by 85%
- Customized LLM workflow for performance and reliability by leveraging asynchronous processing and CUDA integration for parallel execution, improving overall generation time by 60%

United States Air Force | 402 Software Engineering Group

Warner Robins, GA

Software Engineering Intern

June 2024 - August 2024

- Engineered a YOLOv8-based computer vision model via Roboflow, enabling autonomous vehicle navigation across diverse terrains, achieving a 41% mAP (mean Average Precision) for real-time object detection
- Orchestrated the deployment of the machine learning navigation system on a physical testbed, validating model efficacy with a 90% success rate across a variety of environments
- Optimized model inference via TensorRT INT8 quantization and pruning, achieving a 4x reduction in compute footprint, resolving critical hardware bottlenecks, enabling concurrent execution of vision, mapping, and planning stacks

Projects

PropSage | Real-Time Sports Analytics Platform

TypeScript, React.js, Next.js, Node.js, Express.js, Python, Docker

- Designed and developed a full-stack real-time sports analytics platform, implementing Monte Carlo simulations with 50,000+ iterations for fair line pricing and integrating AI APIs for real-time video and news analysis
- Architected a scalable monorepo with Turborepo and pnpm, establishing CI/CD pipeline using GitHub Actions, Docker, and Playwright for testing and deployment across environments
- Built a modern, responsive user interface with React, Framer Motion, and Tailwind CSS, optimizing performance and data fetching using SWR, React Query, and LRU caching to deliver a fast and reliable user experience

Qwyse | AI Resume Assistant

Express.js, React.js, Node.js, Azure AI Studio, SQL

- Managed a team of five engineers throughout the architectural design and full stack development of an AI-powered resume assistant, overseeing transition from initial prototype to a production launch serving 100+ active users
- Implemented a full-context LLM generation system using Azure AI Studio Prompt Flow to gather and synthesize a user's complete history, improving users' ATS-graded resume scores by an average of 20 points
- Spearheaded full-stack integration (React, Node.js/Express.js, SQL) with Azure Cloud Services, establishing robust APIs for data persistence and retrieval, achieving a 95% user satisfaction rate

Skills

Languages: Python, Java, Javascript, Typescript, C, Assembly, SQL, HTML/CSS

Frameworks & Libraries: React.js, Node.js, Express.js, TensorFlow, Oauth, Pandas, Firebase, Tailwind, Flask, FastAPI

Cloud & Tools: Git, Docker, Jupyter, Figma, Microsoft Azure, MySQL, Linux/Unix, Android Studio, AWS, Jest